

Technical Appendix: Sharpe-ratio variance inflation in entity-time financial panels

Dhananjay S. Chauhan

June 1, 2026

1 Purpose

This appendix contains proof details and secondary empirical artifacts that are omitted from the journal-facing main manuscript for length. The main manuscript is the claim boundary; this appendix is the audit trail for derivations, positive controls, robustness tables, and generated public-data artifacts.

2 Mean-Return Design-Effect Proof

For a selected row model $y_{t,i} = \mu + \lambda_t + \epsilon_{t,i}$ with constant selected count $\bar{n}$, same-date intraclass correlation ρ , and independent dates, the row-pooled mean equals the average of date means:

$$\hat{\mu} = \frac{1}{T\bar{n}} \sum_{t=1}^T \sum_{i=1}^{\bar{n}} y_{t,i} = \frac{1}{T} \sum_{t=1}^T \bar{y}_t.$$

The row-naive variance report is $\sigma^2/(T\bar{n})$. The true one-date variance is

$$\text{Var}(\bar{y}_t) = \frac{1}{\bar{n}^2} \{ \bar{n}\sigma^2 + \bar{n}(\bar{n}-1)\rho\sigma^2 \} = \frac{\sigma^2}{\bar{n}} \{ 1 + (\bar{n}-1)\rho \}.$$

Thus

$$\frac{se_{date,true}}{se_{row,naive}} = \sqrt{1 + (\bar{n}-1)\rho}.$$

This is the Moulton-like design effect used as the mean-return special case in the main text.

3 Sharpe UVIF Derivation

Let R_t be the date-level portfolio excess return and $G_t = (R_t, R_t^2)'$. The annualized Sharpe ratio is the smooth functional

$$f(\mu, m_2) = \sqrt{252} \frac{\mu}{(m_2 - \mu^2)^{1/2}}, \quad d_R = \nabla f(\mu_R, m_{2,R}).$$

On sets where $v_R = m_{2,R} - \mu_R^2 > 0$, the first-order expansion is

$$\widehat{\text{SR}}_R - \text{SR}_R = d'_R(\hat{\theta}_R - \theta_R) + o_p(T_R^{-1/2}), \quad \theta_R = \mathbb{E}[G_t].$$

With long-run covariance $\Omega_R = \Gamma_0^R + \sum_{k \geq 1} (\Gamma_k^R + \Gamma_k^{R'})$,

$$\text{Var}_{LR}(\widehat{\text{SR}}_R) = T_R^{-1} d'_R \Omega_R d_R + o(T_R^{-1}).$$

The row-pooled IID calculation applies the same delta method to $\bar{n}T_R$ row observations $H_{t,i} = (Y_{t,i}, Y_{t,i}^2)'$:

$$\text{Var}_{IID, \text{row}}(\widehat{\text{SR}}_{\text{row}}) = (\bar{n}T_R)^{-1} d'_Y \Psi_Y d_Y + o((\bar{n}T_R)^{-1}).$$

Taking the ratio gives the Sharpe UVIF:

$$\text{UVIF}_{\text{SR}} = \frac{\bar{n} d'_R \Omega_R d_R}{d'_Y \Psi_Y d_Y} = \frac{\bar{n} d'_R \Gamma_0^R d_R}{d'_Y \Psi_Y d_Y} \times \frac{d'_R \Omega_R d_R}{d'_R \Gamma_0^R d_R}.$$

The first term is the cross-sectional component and the second term is the serial long-run factor. Componentwise,

$$d'_R \Omega_R d_R = \frac{252}{v_R^3} \left[m_{2,R}^2 \Omega_{11} - \frac{\mu_R m_{2,R}}{2} (\Omega_{12} + \Omega_{21}) + \frac{\mu_R^2}{4} \Omega_{22} \right].$$

The Ω_{22} term contains long-run covariances of squared returns and therefore brings fourth moments into Sharpe-ratio variance.

4 Capacity stress extension

The public sorted-portfolio examples in the main manuscript do not contain average daily volume or order-size information, so they do not support a capacity claim. For traded-security panels where those inputs are available, a square-root impact diagnostic can be added as an extension to the statistical claim boundary.

Proposition 1 (Asymptotic capacity frontier under square-root impact). *Suppose a traded panel has row-pooled gross mean $\bar{r}_{\text{gross}}$, row-pooled return standard deviation σ_{row} ,*

average turnover $\widehat{TO} > 0$, T trading dates, square-root impact cost $\lambda\widehat{TO}\sqrt{A/V}$, and a dependence standard-error inflation factor $\mathcal{M} \geq 1$ relative to row-pooled inference. For a one-sided level- α viability margin, any statistically certified capacity A^* must satisfy

$$A^* \leq V \left(\frac{\bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T}}{\lambda\widehat{TO}} \right)^2 \quad (1)$$

when the numerator is positive; otherwise no positive capacity is certified by the margin.

For a square-root impact diagnostic, the dependence-adjusted one-sided lower confidence margin for gross mean return is

$$\bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T},$$

where σ_{row} is the standard deviation of row-pooled returns before date aggregation or UVIF adjustment. Square-root impact subtracts $\lambda\widehat{TO}\sqrt{A/V}$. Requiring the net lower margin to be nonnegative gives

$$\lambda\widehat{TO}\sqrt{A/V} \leq \bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T}.$$

Rearranging yields the capacity frontier in Proposition 1. If the volatility input is already the date-level portfolio volatility, the date boundary has already embedded the cross-sectional UVIF component, so the cross-sectional part of $\mathcal{M}$ must not be multiplied in again.

5 Higher-Moment UVIF and ERIR Details

For $\theta^{(4)} = (\mu, m_2, m_3, m_4)'$, define skewness and kurtosis from the central moments implied by $\theta^{(4)}$. Any adjusted Sharpe functional $h(\theta^{(4)})$ has a delta-method variance $\nabla h' \Omega^{(4)} \nabla h$, where $\Omega^{(4)}$ is the long-run covariance matrix of $(R_t, R_t^2, R_t^3, R_t^4)'$. The higher-moment UVIF is the corresponding ratio of date-level long-run variance to row-pooled IID variance. It is a diagnostic extension, not an additional empirical rejection gate.

The Effective Row Information Retention (ERIR) diagnostic is a design-effect retention calculation, not a hypothesis test or a formal information claim. Under the same equicorrelated approximation used for the mean-return design effect,

$$DE_{\mu, xs} = 1 + (\bar{n} - 1)\rho, \quad \text{ERIR} = \{1 + (\bar{n} - 1)\rho\}^{-1}.$$

Thus an ERIR of 0.111 means that roughly $1 - 0.111 = 0.889$, or 89 percent, of nominal row-level information is redundant under this diagnostic approximation. Confirmatory inference remains based on date-level HAC-delta and dependent-bootstrap procedures.

Table 1: Synthetic positive-control audit.

Panel	$\bar{\rho}$	Adj.	SR	HAC p^+	RW p	Perm. p^+
Synthetic control	0.202	4.58	1.744	7.67e-13	5.00e-04	9.99e-04

Table 2: Loaded registry-defined public candidate evaluation campaign.

Candidate	status	SR	HAC p+	Boot p+	RW p	Perm p	Net SR	Pass 5%
French momentum deciles, rank-threshold panel	Available	0.447	7.54e-05	2.00e-04	2.00e-04	1.000	0.447	no
French WML decile benchmark	Available	0.497	5.53e-06	2.00e-04	2.00e-04	1.00e-04	0.497	yes
French 25 Size/BM, skipped-momentum panel	Available	0.036	0.445	0.458	0.516	0.947	-0.072	no
AQR Betting Against Beta, equity factors	Available	0.741	3.08e-13	2.00e-04	2.00e-04		0.741	no
AQR Quality Minus Junk	Available	0.536	2.26e-05	2.00e-04	2.00e-04		0.536	no
AQR HML Devil	Available	0.336	0.004	0.002	0.002		0.336	no
AQR Value and Momentum Everywhere	Available	2.623	1.62e-04	2.00e-04	2.00e-04		2.623	no

6 Synthetic Positive Control

7 Full Empirical Artifacts

The inferential tables in this appendix report only public sources that loaded and produced reproducible artifacts. Failed source-access attempts remain in the generated provenance files and campaign metadata, but are not treated as empirical evidence.

8 Full Monte Carlo Artifacts

9 Robustness and Registry Notes

The public registry is frozen in `code/campaign_registry.json`. It records loaders, public source metadata, threshold menus, primary thresholds, and lookback/skip parameters. Changing a source, lookback, threshold, cost assumption, or eligibility rule after seeing results creates a new researcher menu and must be reported as exploratory.

Stationarity diagnostics, holdout splits, HAC bandwidth sweeps, prewhitened HAC, fixed-fraction HAC, moving-block bootstrap, and stationary bootstrap results are secondary robustness checks. They are used to identify fragile claims, not to replace the main date-return decision boundary.

Table 3: Panel candidates subject to the full audit gate.

Panel candidate	SR	HAC p+	Boot p+	RW p	Perm p	Net SR	Pass 5%
French momentum deciles, rank-threshold panel	0.447	7.54e-05	2.00e-04	2.00e-04	1.000	0.447	no
French WML decile benchmark	0.497	5.53e-06	2.00e-04	2.00e-04	1.00e-04	0.497	yes
French 25 Size/BM, skipped-momentum panel	0.036	0.445	0.458	0.516	0.947	-0.072	no

Table 4: Single-series factor benchmarks subject to time-series inference only.

Single-series benchmark	SR	HAC p+	Boot p+	RW p	Scope
AQR Betting Against Beta, equity factors	0.741	3.08e-13	2.00e-04	2.00e-04	time-series only
AQR Quality Minus Junk	0.536	2.26e-05	2.00e-04	2.00e-04	time-series only
AQR HML Devil	0.336	0.004	0.002	0.002	time-series only
AQR Value and Momentum Everywhere	2.623	1.62e-04	2.00e-04	2.00e-04	time-series only

Table 5: Canonical French momentum benchmark validation.

Candidate	thr	n	start	end	panel SR	direct WML SR	scale
French WML decile benchmark	0.900	26131	1926-11-04	2026-04-30	0.497	0.497	0.500

Table 6: Sampling-boundary design-effect diagnostics.

Candidate	status	thr	rows	dates	avg names	rho	SE infl.	row t	HAC z	row p+	HAC p+	RW p
AQR Betting Against Beta, equity factors	not_applicable_single_series	0.000	25281	25281	1.000							2.00e-04
AQR HML Devil	not_applicable_single_series	0.000	26590	26590	1.000							0.002
AQR Quality Minus Junk	not_applicable_single_series	0.000	17681	17681	1.000							2.00e-04
AQR Value and Momentum Everywhere	not_applicable_single_series	0.000	651	651	1.000							2.00e-04
French momentum deciles, rank-threshold panel	Available	0.700	104524	26131	4.000	-0.138	0.766	3.584	3.790	1.70e-04	7.54e-05	2.00e-04
French WML decile benchmark	Available	0.900	52262	26131	2.000	-0.583	0.646	3.299	4.395	4.85e-04	5.53e-06	2.00e-04
French 25 Size/BM, skipped-momentum panel	Available	0.500	45201	3477	13.000	0.667	3.000	0.664	0.222	0.253	0.412	0.516

Table 7: Dependence audit summary for panel candidates.

Candidate	Row p+	UVIF	ERIR	Audit p+	Net SR	Status
French momentum deciles, rank-threshold panel	1.70e-04	1.000	1.000	1.000	0.447	Fails permutation gate
French WML decile benchmark	4.85e-04	1.000	1.000	2.00e-04	0.497	Passes composite gate
French 25 Size/BM, skipped-momentum panel	0.253	9.002	0.111	0.947	-0.072	Fails cost gate

Note: ERIR is a design-effect retention diagnostic and is not used as a rejection rule.

Table 8: Horizon-effect audit for the dynamic Size/BM momentum panel.

L	thr	dates	avg names	gross SR	net SR 5bps	turnover	UVIF SR	serial UVIF	HAC p+
1	0.500	3497	12.913	0.073	-0.394	0.733	11.498	0.891	0.386
5	0.500	3493	13.000	-0.312	-0.528	0.339	12.880	0.990	0.878
21	0.500	3477	13.000	0.058	-0.050	0.169	12.121	0.933	0.412

Table 9: Primary-threshold dependence-aware Sharpe inference.

Candidate	thr	Method	SR	SE	CI low	CI high	p+	n
AQR Betting Against Beta, equity factors	0.000	block	0.741	0.106	0.533	0.951	2.00e-04	25281
AQR Betting Against Beta, equity factors	0.000	HAC	0.741	0.103	0.540	0.943	3.08e-13	25281
AQR Betting Against Beta, equity factors	0.000	prewhite	0.741	0.104	0.538	0.945	4.38e-13	25281
AQR HML Devil	0.000	block	0.336	0.114	0.105	0.552	0.002	26590
AQR HML Devil	0.000	HAC	0.336	0.126	0.089	0.583	0.004	26590
AQR HML Devil	0.000	prewhite	0.336	0.111	0.119	0.553	0.001	26590
AQR Quality Minus Junk	0.000	block	0.536	0.133	0.276	0.795	2.00e-04	17681
AQR Quality Minus Junk	0.000	HAC	0.536	0.131	0.279	0.794	2.26e-05	17681
AQR Quality Minus Junk	0.000	prewhite	0.536	0.128	0.285	0.787	1.40e-05	17681
AQR Value and Momentum Everywhere	0.000	block	2.623	0.716	1.230	4.041	2.00e-04	651
AQR Value and Momentum Everywhere	0.000	HAC	2.625	0.730	1.194	4.056	1.62e-04	651
AQR Value and Momentum Everywhere	0.000	prewhite	2.625	0.723	1.207	4.043	1.42e-04	651
French momentum deciles, rank-threshold panel	0.700	block	0.447	0.114	0.222	0.667	2.00e-04	26131
French momentum deciles, rank-threshold panel	0.700	HAC	0.447	0.118	0.216	0.679	7.54e-05	26131
French momentum deciles, rank-threshold panel	0.700	prewhite	0.447	0.114	0.224	0.670	4.18e-05	26131
French WML decile benchmark	0.900	block	0.497	0.111	0.275	0.708	2.00e-04	26131
French WML decile benchmark	0.900	HAC	0.497	0.113	0.275	0.718	5.53e-06	26131
French WML decile benchmark	0.900	prewhite	0.497	0.111	0.279	0.715	3.99e-06	26131
French 25 Size/BM, skipped-momentum panel	0.500	block	0.036	0.261	-0.487	0.530	0.458	3477
French 25 Size/BM, skipped-momentum panel	0.500	HAC	0.036	0.260	-0.473	0.544	0.445	3477

Table 10: Same-date signal-permutation diagnostics for panel candidates.

Candidate	thr	obs SR	null mean	null sd	p+	perms
French momentum deciles, rank-threshold panel	0.000	0.395	0.410	0.039	0.646	10000
French momentum deciles, rank-threshold panel	0.300	0.419	0.434	0.024	0.733	10000
French momentum deciles, rank-threshold panel	0.500	0.419	0.434	0.024	0.733	10000
French momentum deciles, rank-threshold panel	0.700	0.447	0.447	1.11e-16	1.000	10000
French momentum deciles, rank-threshold panel	0.900	0.497	0.404	0.042	0.015	10000
French WML decile benchmark	0.900	0.497	-0.001	0.098	1.00e-04	10000
French 25 Size/BM, skipped-momentum panel	0.000	0.011	0.052	0.017	0.993	10000
French 25 Size/BM, skipped-momentum panel	0.300	0.010	0.072	0.023	0.996	10000
French 25 Size/BM, skipped-momentum panel	0.500	0.036	0.083	0.029	0.947	10000
French 25 Size/BM, skipped-momentum panel	0.700	0.061	0.112	0.040	0.893	10000
French 25 Size/BM, skipped-momentum panel	0.900	0.025	0.098	0.069	0.855	10000

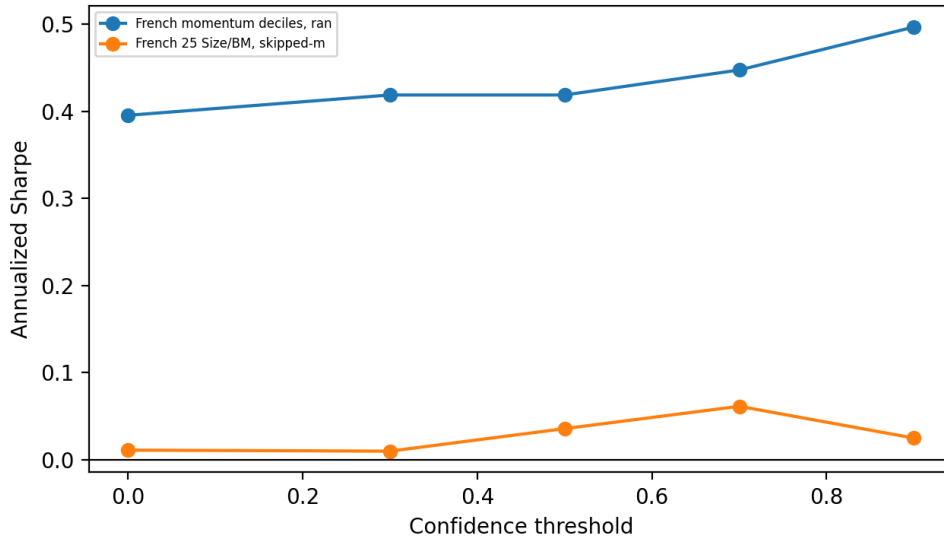


Figure 1: Threshold profile for public candidate panels.

Table 11: HAC bandwidth, prewhitening, and fixed-b robustness diagnostics.

Candidate	Diagnostic	SR	SE	p+
AQR Betting Against Beta, equity factors	HAC K=auto	0.741	0.103	3.08e-13
AQR Betting Against Beta, equity factors	HAC K=21	0.741	0.119	2.04e-10
AQR Betting Against Beta, equity factors	HAC K=63	0.741	0.130	6.06e-09
AQR Betting Against Beta, equity factors	prewhite	0.741	0.104	4.38e-13
AQR Betting Against Beta, equity factors	fixed-b 0.1	0.741	0.166	0.002
AQR HML Devil	HAC K=auto	0.336	0.126	0.004
AQR HML Devil	HAC K=21	0.336	0.128	0.004
AQR HML Devil	HAC K=63	0.336	0.137	0.007
AQR HML Devil	prewhite	0.336	0.111	0.001
AQR HML Devil	fixed-b 0.1	0.336	0.110	0.004
AQR Quality Minus Junk	HAC K=auto	0.536	0.131	2.26e-05
AQR Quality Minus Junk	HAC K=21	0.536	0.140	6.33e-05
AQR Quality Minus Junk	HAC K=63	0.536	0.147	1.32e-04
AQR Quality Minus Junk	prewhite	0.536	0.128	1.40e-05
AQR Quality Minus Junk	fixed-b 0.1	0.536	0.151	0.002
AQR Value and Momentum Everywhere	HAC K=auto	2.625	0.730	1.62e-04
AQR Value and Momentum Everywhere	HAC K=21	2.625	0.712	1.14e-04
AQR Value and Momentum Everywhere	HAC K=63	2.625	0.659	3.41e-05
AQR Value and Momentum Everywhere	prewhite	2.625	0.723	1.42e-04
AQR Value and Momentum Everywhere	fixed-b 0.1	2.625	0.656	0.004
French momentum deciles, rank-threshold panel	HAC K=auto	0.447	0.118	7.54e-05
French momentum deciles, rank-threshold panel	HAC K=21	0.447	0.120	9.94e-05
French momentum deciles, rank-threshold panel	HAC K=63	0.447	0.122	1.21e-04
French momentum deciles, rank-threshold panel	prewhite	0.447	0.114	4.18e-05
French momentum deciles, rank-threshold panel	fixed-b 0.1	0.447	0.143	0.004
French WML decile benchmark	HAC K=auto	0.497	0.113	5.53e-06
French WML decile benchmark	HAC K=21	0.497	0.116	9.62e-06
French WML decile benchmark	HAC K=63	0.497	0.117	1.01e-05
French WML decile benchmark	prewhite	0.497	0.111	3.99e-06
French WML decile benchmark	fixed-b 0.1	0.497	0.147	0.002

Table 12: Composite gate sensitivity by alpha threshold.

alpha	candidates	passes
0.010	8	1
0.050	8	1

Table 13: Holdout and subperiod Sharpe diagnostics.

Candidate	window	SR	n
AQR Betting Against Beta, equity factors	train_70pct	0.887	17696
AQR Betting Against Beta, equity factors	holdout_30pct	0.531	7585
AQR Betting Against Beta, equity factors	quarter_min	0.280	6320
AQR Betting Against Beta, equity factors	quarter_max	1.270	6320
AQR HML Devil	train_70pct	0.465	18613
AQR HML Devil	holdout_30pct	0.120	7977
AQR HML Devil	quarter_min	0.273	6647
AQR HML Devil	quarter_max	0.445	6647
AQR Quality Minus Junk	train_70pct	0.794	12376
AQR Quality Minus Junk	holdout_30pct	0.305	5305
AQR Quality Minus Junk	quarter_min	0.174	4421
AQR Quality Minus Junk	quarter_max	1.083	4420
AQR Value and Momentum Everywhere	train_70pct	2.867	455
AQR Value and Momentum Everywhere	holdout_30pct	1.974	196
AQR Value and Momentum Everywhere	quarter_min	1.707	163
AQR Value and Momentum Everywhere	quarter_max	4.313	163
French momentum deciles, rank-threshold panel	train_70pct	0.615	18292
French momentum deciles, rank-threshold panel	holdout_30pct	0.229	7839
French momentum deciles, rank-threshold panel	quarter_min	0.171	6532
French momentum deciles, rank-threshold panel	quarter_max	1.280	6533
French WML decile benchmark	train_70pct	0.641	18292
French WML decile benchmark	holdout_30pct	0.310	7839
French WML decile benchmark	quarter_min	0.224	6533
French WML decile benchmark	quarter_max	1.370	6533
French 25 Size/BM, skipped-momentum panel	train_70pct	0.050	2428
French 25 Size/BM, skipped-momentum panel	holdout_30pct	0.005	1049
French 25 Size/BM, skipped-momentum panel	quarter_min	-0.657	853
French 25 Size/BM, skipped-momentum panel	quarter_max	0.211	875

Table 14: Turnover-scaled cost sensitivity and break-even cost.

Candidate	cost bps	turnover	gross SR	net SR	break-even bps
AQR Betting Against Beta, equity factors	0	0.000	0.741	0.741	
AQR Betting Against Beta, equity factors	1	0.000	0.741	0.741	
AQR Betting Against Beta, equity factors	5	0.000	0.741	0.741	
AQR Betting Against Beta, equity factors	10	0.000	0.741	0.741	
AQR Betting Against Beta, equity factors	25	0.000	0.741	0.741	
AQR HML Devil	0	0.000	0.336	0.336	
AQR HML Devil	1	0.000	0.336	0.336	
AQR HML Devil	5	0.000	0.336	0.336	
AQR HML Devil	10	0.000	0.336	0.336	
AQR HML Devil	25	0.000	0.336	0.336	
AQR Quality Minus Junk	0	0.000	0.536	0.536	
AQR Quality Minus Junk	1	0.000	0.536	0.536	
AQR Quality Minus Junk	5	0.000	0.536	0.536	
AQR Quality Minus Junk	10	0.000	0.536	0.536	
AQR Quality Minus Junk	25	0.000	0.536	0.536	
AQR Value and Momentum Everywhere	0	0.000	2.623	2.623	
AQR Value and Momentum Everywhere	1	0.000	2.623	2.623	
AQR Value and Momentum Everywhere	5	0.000	2.623	2.623	
AQR Value and Momentum Everywhere	10	0.000	2.623	2.623	
AQR Value and Momentum Everywhere	25	0.000	2.623	2.623	
French momentum deciles, rank-threshold panel	0	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	1	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	5	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	10	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	25	1.91e-05	0.447	0.447	85893.000
French WML decile benchmark	0	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	1	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	5	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	10	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	25	1.91e-05	0.497	0.497	122232.000
French 25 Size/BM, skipped-momentum panel	0	0.169	0.036	0.036	
French 25 Size/BM, skipped-momentum panel	1	0.169	0.036	0.014	
French 25 Size/BM, skipped-momentum panel	5	0.169	0.036	-0.072	
French 25 Size/BM, skipped-momentum panel	10	0.169	0.036	-0.179	
French 25 Size/BM, skipped-momentum panel	25	0.169	0.036	-0.502	

Table 15: Simulation DGP parameter grid.

DGP	Name	T	N	mu	sigma	AR1	rho	GARCH a	GARCH b	K	regime
01_iid	IID baseline	1000	50	4.00e-04	0.015	0.000	0.000	0.000	0.000	1	no
02_ser_mild	Serial mild	1000	50	4.00e-04	0.015	0.200	0.000	0.000	0.000	1	no
03_ser_strong	Serial strong	1000	50	4.00e-04	0.015	0.500	0.000	0.000	0.000	1	no
04_xs_mild	Cross mild	1000	50	4.00e-04	0.015	0.000	0.200	0.000	0.000	1	no
05_xs_strong	Cross strong	1000	50	4.00e-04	0.015	0.000	0.500	0.000	0.000	1	no
06_both_mod	Both moderate	1000	50	4.00e-04	0.015	0.200	0.200	0.000	0.000	1	no
07_both_strong	Both strong	1000	50	4.00e-04	0.015	0.500	0.500	0.000	0.000	1	no
08_realistic	Realistic	2500	100	4.00e-04	0.015	0.150	0.350	0.000	0.000	1	no
09_garch_mild	GARCH mild	1000	50	4.00e-04	0.015	0.200	0.200	0.050	0.900	1	no
10_garch_strong	GARCH strong	1000	50	4.00e-04	0.015	0.300	0.400	0.100	0.850	1	no
11_multifactor	Multi-factor K=3	1000	50	4.00e-04	0.015	0.200	0.300	0.000	0.000	3	no
12_regime	Regime-switching	2000	50	4.00e-04	0.015	0.150	0.250	0.000	0.000	1	yes
13_high_ar1	High-persistence AR(1)	1000	50	4.00e-04	0.015	0.700	0.300	0.000	0.000	1	no
14_garch_highvol	High-volatility GARCH	1000	50	4.00e-04	0.015	0.200	0.300	0.150	0.800	1	no

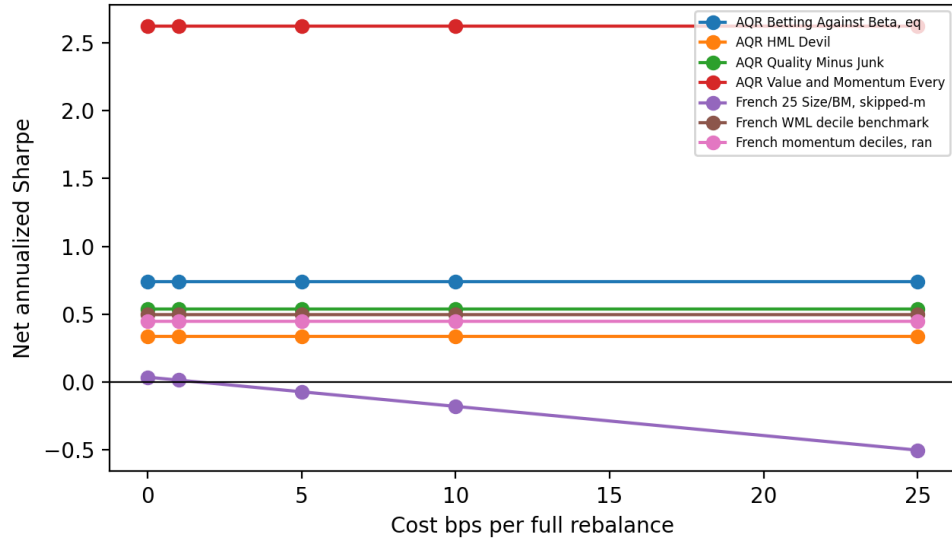


Figure 2: Turnover-scaled cost sensitivity at primary candidate thresholds.

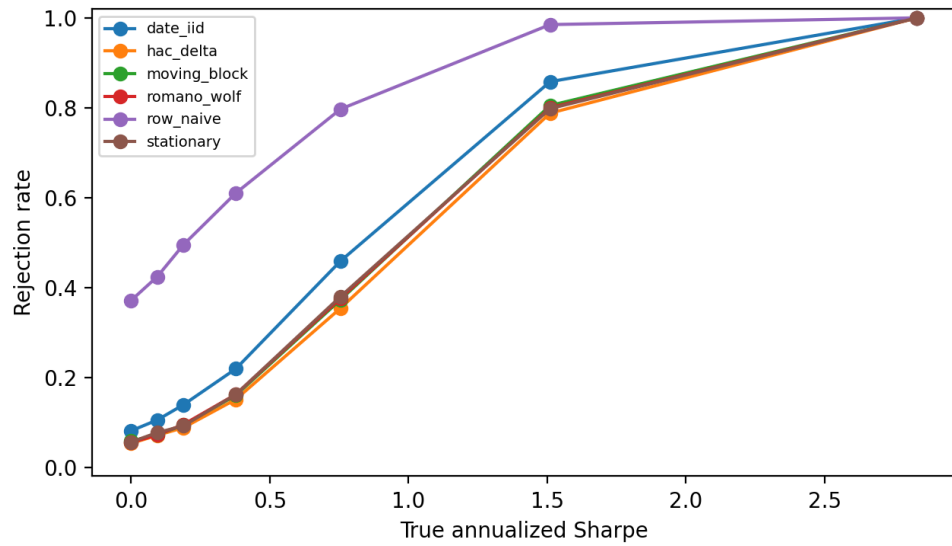


Figure 3: Monte Carlo rejection rates by true annualized Sharpe.

Table 16: Monte Carlo 95 percent interval coverage for the date-level Sharpe target.

DGP	blocked	hac_delta	iid	row_naive	stationary
01_iid	0.947	0.952	0.951	0.000	0.947
02_ser_mild	0.938	0.943	0.898	0.000	0.936
03_ser_strong	0.921	0.934	0.734	0.000	0.919
04_xs_mild	0.950	0.950	0.952	0.073	0.950
05_xs_strong	0.934	0.938	0.942	0.262	0.938
06_both_mod	0.937	0.939	0.898	0.088	0.936
07_both_strong	0.934	0.943	0.771	0.178	0.935
08_realistic	0.937	0.941	0.901	0.107	0.937
09_garch_mild	0.926	0.940	0.877	0.110	0.930
10_garch_strong	0.926	0.948	0.862	0.199	0.927
11_multifactor	0.937	0.947	0.895	0.225	0.940
12_regime	0.929	0.934	0.890	0.306	0.929
13_high_ar1	0.899	0.933	0.562	0.137	0.907
14_garch_highvol	0.940	0.948	0.895	0.193	0.941

Table 17: Null rejection rates at nominal 5 percent size.

dgp	date_iid	hac_delta	moving_block	romano_wolf	row_naive	stationary
null_dep	0.094	0.066	0.070	0.068	0.352	0.067
null_garch	0.098	0.063	0.070	0.067	0.351	0.068
null_iid	0.060	0.060	0.061	0.060	0.062	0.062

Table 18: Rejection rates across true annualized Sharpe values.

index	true_sharpe	date_iid	hac_delta	moving_block	romano_wolf	row_naive	stationary
0.000	0.000	0.081	0.053	0.058	0.055	0.371	0.056
1.000	0.094	0.105	0.071	0.072	0.072	0.424	0.077
2.000	0.189	0.139	0.088	0.094	0.095	0.495	0.092
3.000	0.378	0.219	0.151	0.160	0.162	0.610	0.162
4.000	0.755	0.459	0.354	0.373	0.376	0.797	0.380
5.000	1.511	0.858	0.788	0.805	0.800	0.985	0.798
6.000	2.833	1.000	1.000	1.000	1.000	1.000	1.000